WORKSHEET

Chapter 10: Simple interval estimates

1 From a group of 60 people at a shopping centre, 9 said they did not use a mobile phone. Estimate the probability that an Australian person does not use a mobile phone, to within one standard deviation.

2 From a class of 32 Year-12 students, 21 stated that they spent more than 15 hours a week using social media. Use this information to estimate the probability that a Year-12 student spends more than 15 hours a week using social media, to within 1.5 standard deviations.

3 From a group of 38 students at a large high school, 7 said they lived in a one-parent household. What is the probability that an Australian student comes from a one-parent household, to within two standard deviations.

4 From a group of 54 people waiting to catch a bus in the Brisbane CBD, 22 said they regularly used Facebook for social networking. Estimate the probability that Australians use Facebook, to within 1.8 standard deviations.

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5 A group of 35 parents attending a high school parent–teacher evening were asked if they thought that teenagers spend too much time using a computer. Nineteen of them agreed. What is the probability that parents of high school students think teenagers spend too much time using a computer, to within 1.6 standard deviations?

6 Forty-four people at a Melbourne fast-food outlet were assessed and it was found that 25 had a body mass index (BMI) measurement of more than 27. What is the probability that Australians have a BMI greater than 27, to within 1.2 standard deviations?

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Answers

1 Between 0.104 and 0.196

2 Between 0.530 and 0.782

3 Between 0.058 and 0.310

4 Between 0.287 and 0.528

5 Between 0.408 and 0.678

6 Between 0.479 and 0.658

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